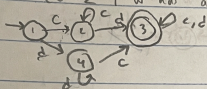


① Give state Diagrams

Σ^* means ϵ included!
 $\Sigma = \{c, d\}$

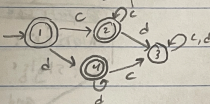
A:

a) $A = \{w \in \Sigma^* \mid w \text{ has at least one } d \text{ and at least one } c\}$



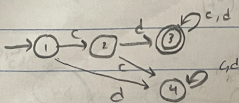
B:

b) $B = \{w \in \Sigma^* \mid \text{either } w \text{ has no } c \text{ in it or } w \text{ has no } d \text{ in it}\}$



C:

c) $C = \{w \in \Sigma^* \mid w \text{ begins with substring } cd\}$

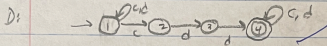


4.5
 4.5

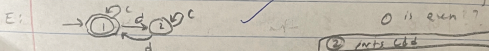
7.5
7.5

2) Give NFAs $\Sigma = \{c, d\}$

a) $D = \{w \in \Sigma^* \mid w \text{ has odd no. of } d\}$

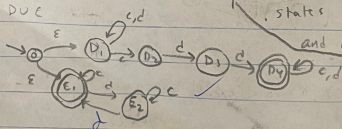


b) $E = \{w \in \Sigma^* \mid w \text{ has an even no. of } d\}$



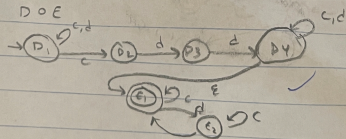
c) DUE

DUE



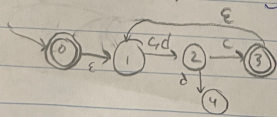
d) DOE

DOE



e) F^* when $F = \{cc, dc\}$

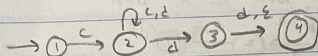
F^*



2 parts odd
NOTICE: I am prefixly
states D as D_n
and E as E_n

For each
 i) give set of states NFA can be in after reading
 ii) whether or not NFA accepts input string

③



a) ccd

ai) $\{2, 3, 4\}$

aii) YES, the NFA accepts "ccd"

b) dc

bi) $\{1\} \rightarrow \emptyset$

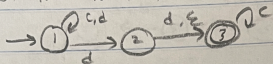
bii) No, the NFA rejects "dc"

c) cdc

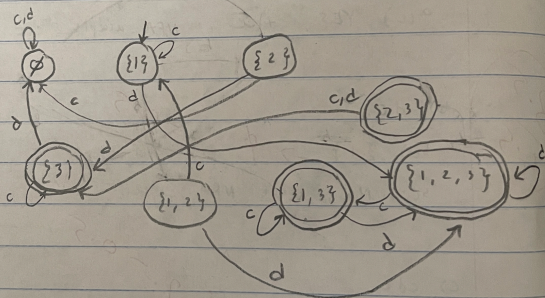
ci) $\{2, 3\} \rightarrow \{2\}$

cii) No, the NFA rejects "cdc"

④ Convert the NFA to a DFA



new states : $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$



3.5
3.5